PROACTIVE–ADAPTIVE PUBLIC TRANSPORT PLANNING: BUS NETWORK REDESIGN FOR URBAN RAIL DISRUPTIONS

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Abstract: Accessibility, the extent to which opportunities can be reached within a given travel-time threshold, is a central benefit of public transport. Urban Rail Transit (URT) disruptions remove much of it, and the usual remedy, replacement buses shuttling between the affected stations, restores the form of rail service rather than access to destinations. This paper proposes a proactive–adaptive framework for redesigning the bus network under URT disruptions. Before any disruption, the operator prepares a few contingency route variants; afterwards, routes and frequencies are adapted in time phases, with prepared variants available soon after the disruption and unprepared ones only after a planning delay. Every variant keeps at least two thirds of its original route segments. The objective is 30-minute cumulative accessibility. The model is a defender–attacker–defender program whose worst-case objective can be blended with the expected value of a two-stage stochastic program. We solve it with an exact column-and-constraint generation algorithm that returns machine-checkable certified bounds. In a protocol-fixed comparison under worst-case one- and two-link rail failures, the benchmark is frequency-optimal bridging: replacement buses on a declared sub-chain menu, released 10 minutes after the disruption. On two synthetic grid networks, prepared rerouting with frequency adjustment and no shuttles recovers 18.9–19.5% of the worst-case accessibility loss, against 12.8–16.4% for bridging, and combining rerouting with the bridging shuttles recovers 22–23%. On a network built from Rennes (France) data, bridging is stronger than rerouting alone (43.3% vs. 17.8–19.1%), and the combination exceeds it by only 1.1–2.3 percentage points. Redesign is therefore not a universal substitute for replacement buses.

Keywords: Proactive-adaptive strategy; Accessibility; Public transport resilience; Bus bridging; Defender–attacker–defender model

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1. Introduction

Urban Rail Transit (URT) systems form the backbone of metropolitan mobility, providing high-capacity, reliable services under normal operations (Tan et al., 2020). Yet this critical infrastructure remains vulnerable to diverse disruptions, from unexpected incidents to planned maintenance: a track fire at New York City's 145th Street station suspended A, B, C and D service for over two hours during the morning peak (Barone & Brown, 2017), the 2024 summer closure of part of Paris's RER B line was described as a historic interruption (Actu.fr, 2024), and even the highly reliable Hong Kong MTR experiences hundreds of disruptions per year (Zhang & Lo, 2020).

The fallout from URT disruptions signifies more than mere delays; it represents a critical failure of access. Although city centers may have redundant routes, suburbs with thinner networks are far more vulnerable: a single disruption there can cut people off from vital destinations such as jobs, schools, or hospitals (Liu

et al., 2024). Traditional metrics like average delay therefore provide an incomplete picture. A more comprehensive metric is *accessibility*, the practical ability to reach important daily destinations within a reasonable time (Hansen, 1959; Miller, 2020).

The go-to remedy, substitute buses along the affected rail corridor (*bus bridging*), tends to recreate the form of service without delivering its function (Currie & Muir, 2017; Pender et al., 2012): replacement buses often suffer from inadequate capacity, longer travel times, and poor reliability compared to the original rail system (Currie & Muir, 2017). The literature has followed two largely separate streams. *Proactive* studies assess and strengthen networks before disruptions occur, through vulnerability and resilience analysis (Cats & Jenelius, 2014; Rodríguez-Núñez & García-Palomares, 2014; Xu & Chopra, 2022). *Reactive* studies optimize operations after a disruption, for example through rapid-transit recovery planning (Cadarso et al., 2015), train rescheduling (Cacchiani et al., 2014), bus bridging route design and dispatching (Jin et al., 2016; van der Hurk et al., 2016) and operational strategies such as stop skipping (Tang et al., 2018); see Gkiotsalitis (2023) for an overview. Stochastic programs exist for transit network design (An & Lo, 2016), but strategic planning and operational disruption management are still mostly treated separately (Bešinović, 2020). URT disruption research has mostly used accessibility to *measure* impacts (Liu et al., 2024; Nalin et al., 2025). Our earlier work optimized it through reactive bus rerouting for a single, known disruption (Guo et al., 2026). This paper extends that idea to preparation under uncertainty, with four contributions:

1. *Preparation as prepared route variants.* Preparation consists of contingency route variants that are planned and briefed in advance, so they can run soon after a disruption, while unprepared variants become available only after a planning delay. This makes the trade-off between preparing and reacting explicit.
2. *Disruptions as an adversary.* Disruptions are modelled as an adversary removing up to two rail links. The resulting defender–attacker–defender (DAD) program (Brown et al., 2006) is blended with the expectation of the corresponding two-stage stochastic program, and a Path Order Preservation Rate (POPR) constraint keeps every variant close to its original line.
3. *Exact, certified solution.* We give an exact column-and-constraint generation algorithm (Zeng & Zhao, 2013) whose results carry machine-checkable bounds.
4. *Protocol-fixed comparison.* We compare the method with frequency-optimal bridging under a protocol fixed before the comparison runs, on two synthetic networks and one built from Rennes data, and report where it does *not* win.

2. Problem Setting

Demand and network. The study area is partitioned into square tiles $s \in \mathcal{S}$ with population shares P_s ($\sum_s P_s = 1$); tile d holds n_d opportunities. The multimodal graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$ has rail stations, existing bus stops and candidate stops as nodes, and rail links $\mathcal{E}^r$, road links usable by buses and walking links as links. A journey combines walking access and egress, waiting ($30/f$ min at frequency f in veh/h), in-vehicle time and walking transfers. Bus lines $\ell \in \mathcal{L}$ run symmetric services on simple paths p_ℓ^0 at nominal frequency f_ℓ^{nom}.

Route variants and POPR. Each line has a finite, restricted (not exhaustive) pool $\mathcal{P}_\ell$ of alternative routes, frozen before the analysis. Let $P = (v_1, \ldots, v_n)$ be the original stop sequence. A segment $\{v_i, v_{i+1}\}$ is preserved by a route P' if P' traverses it in either direction. With K preserved segments, $\text{POPR}(P, P') = K/(n-1)$. This is a discrete, topology-oriented counterpart of the path overlap ratio (Lim & Kim, 2005). Every variant must satisfy POPR $\geq \rho = 2/3$. Because POPR counts adjacencies and not order, the original stops must also appear in their original order; in addition, the run time between the original terminals may grow by at most $\delta_R = 30\%$, and the whole route takes at most $C^{\max} = 60$ min. Besides local detours, the pool contains, for every line end and every eligible transfer target, the

shortest terminal extension. An eligible transfer target is the nearest bus node within a 5-min walk of a rail station. This extension family was added after exploratory runs; the frozen pools are the files `pool_osr1_m1_20.json` of the repository.

Disruptions. A failure unit is one rail link between adjacent stations. The attack set $\mathcal{Y}$ contains all one- and two-link failures ($\Gamma = 2$). For expected values, links fail independently with probability $\pi = 0.1$, renormalised on $\mathcal{Y}$ to weights q_y.

3. Methodology

3.1 Proactive–adaptive decision process

Decisions are taken at three moments:

1. *Preparation* x, before any disruption, selects at most $K = 3$ variants from the pools. A prepared variant is a complete contingency route, ready to run soon after a disruption. We use a single readiness tier, i.e. no partially prepared variants.
2. The *disruption* $y \in \mathcal{Y}$ is revealed.
3. The *response* $z = (z_j)_j$ is chosen per phase j of the horizon $D = 120$ min:
   - $[0, \mu)$ with $\mu = 10$ min: the original network runs without the failed links;
   - $[\mu, \delta)$ with $\delta = 60$ min: each line runs its original route or a *prepared* variant;
   - $[\delta, D]$: each line may run any variant of its pool.

Frequencies are re-chosen in every phase after the first. Replacement shuttles are available from their release time $r_S \in \{\mu, \delta\}$. Each phase is evaluated as a steady state and weighted by its duration, $w_j = |T_j|/D$, without dynamics across phases.

3.2 Accessibility objective

Let $\mathcal{R}_s^{\text{nom}}$ be the destinations reachable from s within $\tau = 30$ min in the intact network, and $\chi_{sd}(z_j, y) \in \{0,1\}$ the 30-minute reachability in phase j. Accessibility is population-weighted and *capped at nominal*: only nominally reachable pairs count, each at most once. The phase and preparation values are

$$\hat{A}(z_j, y) = \sum_s P_s \sum_{d \in \mathcal{R}_s^{\text{nom}}} n_d \, \chi_{sd}(z_j, y), \qquad V(x, y) = \sum_j w_j \max_{z_j \in Z_j(x,y)} \hat{A}(z_j, y), \tag{1}$$

and the model is

$$\max_{x \in X} \Phi_\lambda(x) = \lambda \min_{y \in \mathcal{Y}} V(x, y) + (1 - \lambda) \sum_{y \in \mathcal{Y}} q_y V(x, y). \tag{2}$$

With $\lambda = 1$ it is a DAD program, with preparation chosen against the worst disruption; with $\lambda = 0$ it is the two-stage stochastic program. Preparation only enlarges the response sets $Z_j(x, y)$; it does not harden rail links. The nominal value is A^{nom}. The unmitigated value $H(y)$ is that of the original routes at nominal frequency without shuttles.

3.3 Response constraints

In each phase, $Z_j(x, y)$ requires:

1. *One route and frequency per line.* Each line runs exactly one admissible route at one frequency from the menu $\{2, 3, 4, 6, 8, 10, 12, 15\}$ veh/h (Rennes adds 5, 7.5 and 9), with $f \geq \max(f^{\min}, \theta f_\ell^{\text{nom}})$, $f^{\min} = 3$, $\theta = 2/3$.
2. *Fleet.* $\sum_\ell \lceil 2(1+\beta) R_\ell f_\ell / 60 \rceil$ plus shuttle vehicles $\leq F$, with run time R_ℓ and layover share $\beta = 0.1$.

3. *Operating cost.* Vehicle-km per hour stay within $(1+\varepsilon)$ of the nominal value, $\varepsilon = 0.1$.

4. *Rider protection.* For every rider leg on a line, the detour's extra in-vehicle time plus the extra wait from a lower frequency is at most $\eta = 10$ min. This is a policy tolerance, not a calibrated value; it applies per leg, so a journey with k affected legs may lose up to $k\eta$.

5. *No harm.* No tile pair reachable in the unmitigated state may lose reachability.

Shuttles count towards the fleet and vehicle-km limits but are exempt from the route and rider-protection rules. The recourse is complete, because the original routes at nominal frequency are always feasible.

3.4 Exact solution and certification

The monolithic formulation (A-DAD) enumerates $\mathcal{Y}$ with one recourse copy per attack and phase, using a linearised reachability block. A benchmark against it is pending.

We use a column-and-constraint generation method (C&CG-C) (Zeng & Zhao, 2013). For $\lambda = 1$ it starts with an *initial bounding* phase, which uses the fact that $\max_x V(x,y)$ bounds the optimum for every attack y. Attacks are processed in the order of preparation-independent upper bounds, and a wait-and-see problem (preparation chosen knowing y) is solved for each; the smallest bound is a valid upper bound. Each wait-and-see preparation is a candidate, and its value against all attacks is bounded from below by screening and exact evaluation, which gives the lower bound. Early phases use a configuration and frequency search; single-attack MILPs handle selected bounding and refinement tasks. Recourse values may remain intervals even in optimal runs. If the bounds do not meet, a master problem over x and an epigraph variable iterates over generated attacks with optimistic, lazily refined per-attack configuration columns, and separation adds attacks certified to violate the incumbent master target, refining unresolved intervals as needed. The initial-bounding phase is disabled for $\lambda < 1$, where a single-attack bound does not bound the expected value; the master loop then runs alone. The method is exact and terminates finitely, since X and $\mathcal{Y}$ are finite, but within a time limit it may return only an interval.

An independent checker re-validates every reported certificate. It binds the certificate to the instance, pool, attack-set and code hashes, re-evaluates every witness response, recomputes the lower bound, and re-derives the upper bound from the recorded solver bounds. "Optimal" means a certified relative gap of at most 10^{-4}, and optimality is relative to the frozen pools.

4. Numerical Results

Instances. Table 1 lists the three validation instances.

- *Grids.* The two grids are rule-generated and uncalibrated, and share one generator. Three rail lines (east–west, north–south, diagonal) cross a central station, and feeder, orbital and crosstown bus lines complete the network. Jobs and population decay exponentially with distance from the center.

- *Rennes.* This instance uses the STAR GTFS feed (two metro lines and six trunk bus lines, Tuesday 07:00–09:00), the INSEE 2021 1-km population grid and OpenStreetMap activity locations as opportunities. Speeds and circuity are calibrated on GTFS run times and the OSM pedestrian network. It remains a reduced geometric approximation, without crowding or capacity and without full journey-time validation, so we use it for relative comparisons only.

The worst-case unmitigated loss $\bar{L} = \max_y(A^{\text{nom}} - H(y))$ is 22–24% of A^{nom} on all three instances.

Compared strategies. All strategies share the fleet, vehicle-km, frequency and no-harm rules and the same exact evaluator.

- *Unmitigated* (U): no response.

- *Frequency only* (FO): original routes with re-optimised frequencies.

Table 1: Validation instances.

Instance	Tiles	Rail stations/links	Bus lines	Cand. stops	Fleet F	$\lvert\mathcal{Y}\rvert$	Variants
Grid 10×12	120 (1.5 km)	19 / 18	11	57	113	171	336
Grid 12×14	168 (1.5 km)	23 / 22	12	93	125	253	350
Rennes	108 (1 km)	28 / 28	6	26	78	406	146

- *Bridging* (RB+): original routes plus replacement shuttles, with at most one slot per contiguous sub-chain of every maximal disrupted chain and frequencies optimised per attack. We call it *frequency-optimal bridging on a declared sub-chain menu*. It strengthens conventional replacement buses, which serve the maximal disrupted chain only.
- *Prepared adaptive rerouting* (PA, our method): preparation and the phased response above, *without* shuttles.
- *Hybrid* (HY): PA plus the RB+ shuttles. Its feasible set contains that of RB+.

Shuttle stops. Shuttles stop at the nearest bus or candidate stop within a 5-min walk of each station, reached by an ordinary transfer of $\max(w, 0.5)$ min for walk time w. Operators typically use kerbside stops near stations, but this exact rule is a modelling choice. On the grids every station has such a stop, 2.6 min away. In the reduced Rennes model four stations have none. There we assume a temporary station-front stop with zero additional transfer time. This is an assumption: its net effect on accessibility has no guaranteed sign.

Performance measure and decision rule. The worst-case recovery of a strategy is

$$\mathrm{WR} = 1 - \frac{A^{\mathrm{nom}} - \min_y V(x,y)}{\bar{L}}, \tag{3}$$

the share of the worst-case unmitigated loss that is recovered; U has $\mathrm{WR} = 0$. Interval results are compared through $\Delta = [A^- - B^+, A^+ - B^-]$. The rule, fixed before the validation comparisons, reads: SUP (better) if $\Delta^- > 0$; SUB (substantially better) if $\Delta^- \geq 0.05$; LOSS if $\Delta^+ < 0$; INC otherwise. HY is judged on SUB only.

Setup. PA and HY were solved with Gurobi 13.0.3 (4 threads) under an 8-GB solver memory limit and an external 10-GB watchdog, on an Intel i5-13400F with 15 GB of RAM, with a 7200-s search budget. Bridging was solved per (attack, shuttle class) unit, using a custom frequency branch-and-bound with exact evaluation and 600 s per unit.

- *Grids.* All six PA/HY runs were certified optimal, with wall times of 753–2426 s, certification of 18–52 s and peak memory of 2.4–3.9 GB (solver) or 2.5–4.6 GB (process tree).
- *Rennes.* PA, HY ($r_S = \mu$) and HY ($r_S = \delta$) reached the time limit. Their wall times were 7250, 7303 and 7347 s; the overruns of 50, 103 and 147 s are spent on the permitted final certification (93, 144 and 179 s). Peak memory was 3.6–3.7 GB (solver) and 4.5–4.7 GB (process tree).
- *Initial bounding.* In eight of the nine runs, initial bounding alone produced the reported bounds, without master iterations.

Per-run status, wall and certification times, both memory peaks, certificates and the generated tables are in the repository https://github.com/zihao-guo/P3_ACC_CASPT_CODE, commit `0774f47`, file `results/dad/bs_validation/tables.md`.

Results. Tables 2–3 report the results. Frequency changes alone recover little (FO: 2.9–10.2%).

Grids: redesign beats bridging. With early shuttles ($r_S = \mu$), PA recovers 18.9–19.5% of the worst-case loss, against 13.1% and at most 16.4% for bridging. On grid 10×12 the advantage is substantial (SUB). On grid 12×14, PA is better (SUP), but SUB is unresolved because 218 of the 506 bridging units ended

Table 2: Worst-case recovery WR (U = 0). PA and HY values are certified intervals (endpoints rounded to four decimals); FO and RB+ values without brackets are exact. Shuttle release $r_S = \mu$ (10 min) is primary, $r_S = \delta$ (60 min) secondary.

	FO	Bridging RB+		PA (ours,	Hybrid HY	
Instance		$r_S = \mu$	$r_S = \delta$	no shuttles)	$r_S = \mu$	$r_S = \delta$
Grid 10×12	0.0293	0.1314	0.0850	[0.1946, 0.1946]	[0.2303, 0.2304]	[0.2141, 0.2141]
Grid 12×14	[0.0286, 0.0803]	[0.1279, 0.1640]	[0.0828, 0.1259]	[0.1891, 0.1891]	[0.2215, 0.2215]	[0.2068, 0.2068]
Rennes	0.1015	0.4327	0.2948	[0.1779, 0.1913]	[0.4438, 0.4554]	[0.1520, 0.3413]

Table 3: Differences Δ to bridging RB+ (same r_S) and verdicts of the fixed decision rule.

Instance	PA, $r_S = \mu$	PA, $r_S = \delta$	HY, $r_S = \mu$	HY, $r_S = \delta$
Grid 10×12	[0.0631, 0.0631] SUB	[0.1096, 0.1096] SUB	[0.0989, 0.0990] SUB	[0.1291, 0.1291] SUB
Grid 12×14	[0.0251, 0.0612] SUP	[0.0631, 0.1063] SUB	[0.0575, 0.0936] SUB	[0.0808, 0.1240] SUB
Rennes	[−0.2548, −0.2414] LOSS	[−0.1169, −0.1035] LOSS	[0.0111, 0.0227] no SUB	[−0.1428, 0.0465] no SUB

with an interval. With late shuttles ($r_S = \delta$), bridging falls to at most 8.5% and 12.6%, and PA is SUB on both grids. The hybrid is SUB on both grids at both release times. We conjecture that a few rerouted lines reach time-critical origin–destination pairs that a station-to-station shuttle on slower roads cannot reach within 30 minutes. Our aggregate results do not test this.

Rennes: bridging is strong. Here bridging with early shuttles recovers 43.3%, while PA without shuttles recovers 17.8–19.1%, a LOSS at both release times. The hybrid exceeds bridging by 1.1–2.3 points with early shuttles, below the SUB threshold. With late shuttles it is inconclusive (no SUB), because its certified interval remains wide after two hours, although its true value cannot be below that of bridging. One untested explanation is that few reroutable bus lines run close to the failed metro links. Shuttle results also depend on the temporary-stop assumption at four stations. The protocol therefore requires a descriptive subset (S-flag): the 171 attacks whose shuttle slots avoid those stations, with the loss $\bar{L}$ recomputed on the subset and the preparations optimized on all 406 attacks re-used without re-optimization. With $r_S = \mu$, bridging recovers 34.9%, PA $\geq$ 12.2% (inconclusive) and HY $\geq$ 35.1% (better, SUB undetermined); with $r_S = \delta$, bridging recovers 23.6%, PA $\geq$ 12.2% and HY $\geq$ 6.6% (both inconclusive). These results are descriptive and not part of the claims.

Claims (primary setting $r_S = \mu$). "PA's worst-case recovery is better than bridging on all three instances" is *not supported* (SUP on two instances, LOSS on one); "HY is substantially better than bridging on all three" is *partially supported*. SUB counts are PA 1/3 and HY 2/3 at $r_S = \mu$, and PA 2/3 and HY 2/3 at $r_S = \delta$. The results are provisional. The protocol also requires the expected-value results ($\lambda = 0$, now running), the rail-node-stop sensitivity (S-rail, needed before the comparative claims are final), the Rennes nearest-stop sensitivity and the no-preparation and conventional-bridging controls. These are queued and not yet run, as is the A-DAD benchmark. Without the no-preparation control, PA's advantage cannot yet be attributed to preparation itself.

Disclosures.

- *Stop rule.* An earlier protocol let shuttles stop at rail nodes with no transfer walk, while buses used kerbside stops. The bus-stop rule was adopted after that choice reversed the ranking on a development grid (metro 8×10), which is excluded from all claims.
- *Prior exposure of the validation instances.* These instances are comparison-unseen only: no PA, HY or RB+ comparison had been run, but unmitigated baselines, rail-node bridging recovery values on the grids, and the Rennes calibration and unmitigated baselines had been seen. The two grids share a generator and test scale, not a different mechanism.
- *Exploratory choices.* The terminal-extension variants, the RB+ menu and four exact solver accelerations were introduced after exploratory runs.

- *Software defects.* Two defects were fixed during the runs. A torn bridging run ledger affected bookkeeping only, not values. A preprocessing memory leak voided three grid 12×14 runs, which were re-run with the same budget.

5. Conclusion and Future Work

This paper presented a proactive–adaptive framework for bus network redesign under URT disruptions. Its objective is 30-minute accessibility, and a POPR constraint keeps every variant close to its line. The model links prepared contingency variants and later ad-hoc rerouting through explicit time phases, and blends the worst case with the expected value. An exact algorithm returns certified bounds.

In a protocol-fixed comparison with early bridging shuttles, prepared rerouting with frequency adjustment recovered more worst-case accessibility than frequency-optimal bridging on two synthetic grids, substantially so on one of them. On the Rennes network, bridging was stronger than rerouting alone, and adding the bridging shuttles to the redesign improved on bridging only marginally. Network redesign is thus not a universal substitute for replacement buses; its value appears to depend on the network.

Future work covers the remaining protocol results, diagnostics of which origin–destination pairs each strategy restores, further real networks, and within-phase dynamics, crowding and congestion.

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